CS231P - Parallel and Distributed Computing

Group 20: Homework 2

OSI Model Exploration and Layered Communications
Analysis via Socket Programming

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1.1 Project Overview and Environment

This assignment involves the use of an isolated virtual network to expose the layered OSI model of communications. Two virtual machines (VMs), TX (client) and RX (server), were configured on a shared network segment using a lightweight Linux distribution. The objective was to follow the packets of communicating processes using a socket API and identify the OSI layers traversed.

1.2 Packet Identification

The packet of interest was captured on the RX machine using Wireshark with a capture filter specific to the VM IP addresses (ip src TX_IP and ip dst RX_IP). This isolated the TCP segment containing the message "Hello Sockets" sent from the TX client.

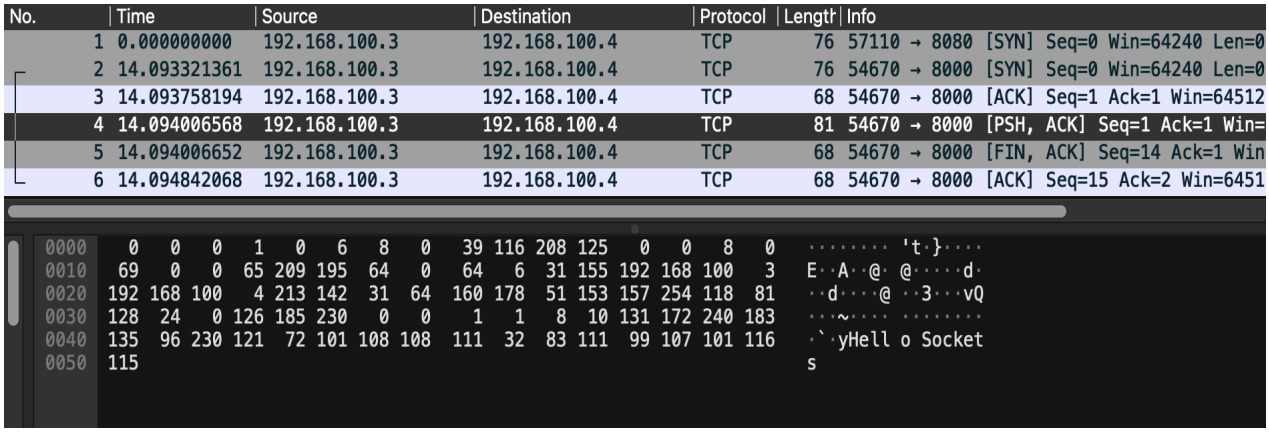
1.3 Detailed Byte-Level Analysis

The following table identifies specific groups of bytes within the captured packet and explains their purpose within the protocol headers:

Bytes Range	Field Name	Purpose/Function
0-5 & 6-11	MAC Addresses	Destination and Source MAC addresses for hardware-level delivery in the local segment.
14 & 16-17	IP Header Info	Byte 14 encodes the Version/Header Length; Bytes 16-17 specify the Total Length of the IP datagram.
18-19	Identification	Used for uniquely identifying fragments of an IP datagram for reassembly.
20-21	Flags/Offset	Controls fragmentation and indicates the position of a fragment in the original packet.
23	Protocol	Identifies the encapsulated protocol (e.g., TCP = 6).
26-29 & 30-33	IP Addresses	Source and Destination IP addresses for logical end-to-end routing (Network Layer).

Bytes Range	Field Name	Purpose/Function
34-35 & 36-37	TCP Ports	Source and Destination ports to identify the specific communicating processes.
After Byte 65	Payload	The actual application data, containing the "Hello Sockets" message.

2.1 Raw Data Screenshot (from “Wireshark”)



No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	192.168.100.3	192.168.100.4	TCP	76	57110 → 8080 [SYN] Seq=0 Win=64240 Len=0
2	14.093321361	192.168.100.3	192.168.100.4	TCP	76	54670 → 8000 [SYN] Seq=0 Win=64240 Len=0
3	14.093758194	192.168.100.3	192.168.100.4	TCP	68	54670 → 8000 [ACK] Seq=1 Ack=1 Win=64512
4	14.094006568	192.168.100.3	192.168.100.4	TCP	81	54670 → 8000 [PSH, ACK] Seq=1 Ack=1 Win=
5	14.094006652	192.168.100.3	192.168.100.4	TCP	68	54670 → 8000 [FIN, ACK] Seq=14 Ack=1 Win
6	14.094842068	192.168.100.3	192.168.100.4	TCP	68	54670 → 8000 [ACK] Seq=15 Ack=2 Win=6451

```
0000   0   0   0   1   0   6   8   0  39 116 208 125   0   0   8   0   ····· 't·}····
0010  69   0   0  65 209 195  64   0  64   6  31 155 192 168 100   3   E··A··@· @·····d·
0020 192 168 100   4 213 142  31  64 160 178  51 153 157 254 118  81   ··d····@ ··3···vQ
0030 128  24   0 126 185 230   0   0   1   1   8  10 131 172 240 183   ···~···· ·····
0040 135  96 230 121  72 101 108 108 111  32  83 111  99 107 101 116   ·`·yHell o Socket
0050 115                                                                 s
```

2.2 OSI Layer Partitioning

The following table associates partitions of the bit block with the respective OSI layers, including the specific byte ranges:

OSI Layer	Byte Range	Description
Layer 2 (Link)	0 - 13	Ethernet header containing MAC addresses and EtherType.
Layer 3 (Network)	14 - 33	IPv4 header containing routing information and logical addresses.
Layer 4 (Transport)	34 - 65	TCP header including ports, sequence numbers, and options.
Layer 7 (Application)	66 - End	The "Hello Sockets" data payload.

2.3 Conclusion

By tracing the packet from hardware addressing (Link Layer) through logical routing (Network Layer) and process-to-process delivery (Transport Layer), we have successfully mapped the encapsulation process used by the OS socket API to transmit data between TX and RX VMs.